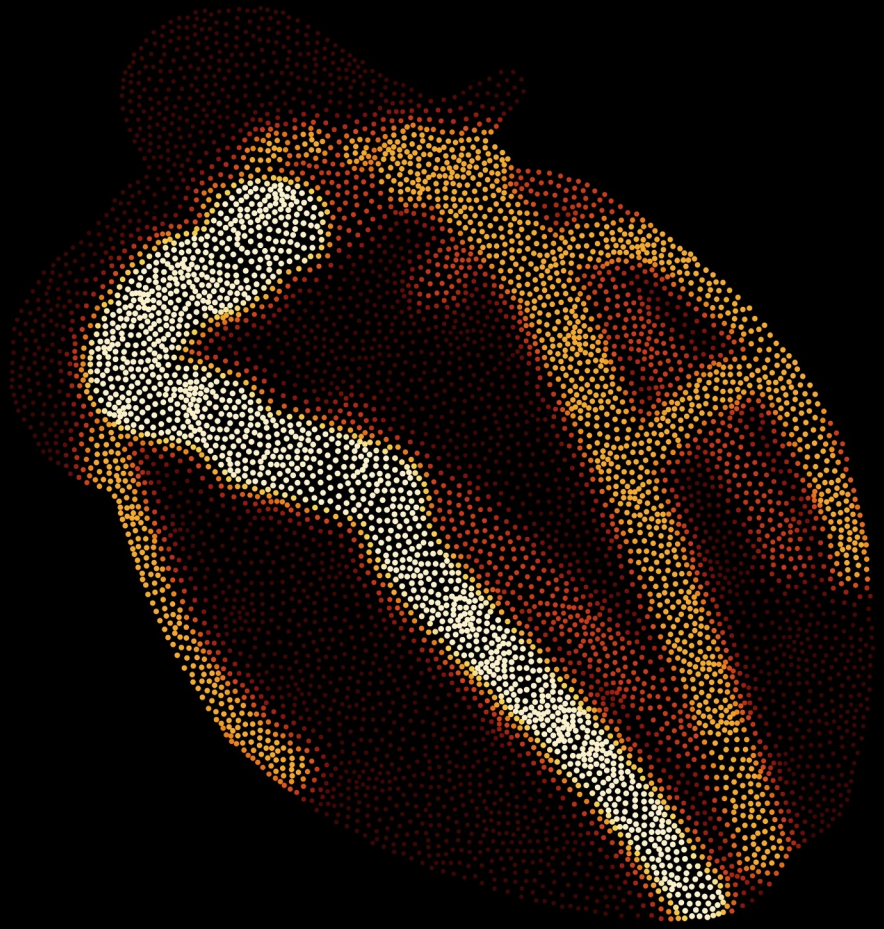
ACC Cardio Sim Creator Competition

CoroSim

A cardiovascular simulator where learners change the cause and watch the effect

corosim.com



A Voronoi stipple of CoroSim's own coronary tree.

One simulated patient, six ways to learn from it

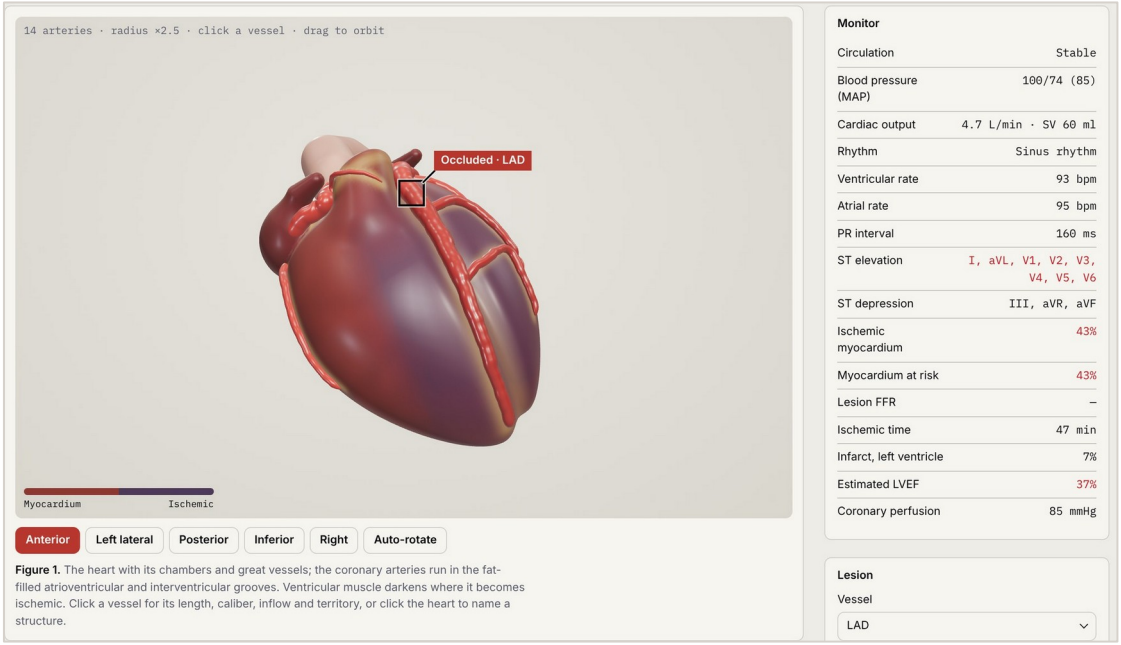


Figure 1. The heart with its chambers and great vessels; the coronary arteries run in the fat-filled atrioventricular and interventricular grooves. Ventricular muscle darkens where it becomes ischemic. Click a vessel for its length, caliber, inflow and territory, or click the heart to name a structure.

Figure 1. Proximal LAD occlusion 47 minutes in: anterior ST elevation, 43% of the myocardium ischemic, LVEF 37%.

Six linked pages

Simulator, daily case, EKG, cath lab, structural and echo, all running the same patient model.

50+ EKG presets

Rhythms, blocks, ischemia and STEMI mimics, plus custom tracings on request.

Free in any browser

No install, account or license.

What this submission answers

Problem solving and efficacy	What students and residents struggle with, and how CoroSim responds	4–12
Skillsets targeted	Six skills, and which pages practice each	13
Intended audience	Medical students and internal medicine residents	14
Assessment metrics and competency evaluation	Three levels, and a planned study in RSOM's Class of 2030	15
Significance and seamless integration	Why it matters, and how it drops into existing teaching	16–17

Learners want more ECG practice than they get

96%

of 401 medical students wanted more ECG training; they rated their teaching 2 of 5, and their confidence in emergencies 2 of 5.

Chaumont et al., BMC Med Educ 2024

83%

of medical students said reading ECGs in a clinical setting makes them anxious.

El-Baba et al., Int J Emerg Med 2023

35%

of answers on core cardiovascular and respiratory concepts stayed wrong from second year to the final year (987 students).

Badenhorst et al., Adv Physiol Educ 2021

No gain

in knowledge after a 14-week ECG lecture series for internal medicine residents, though their confidence rose (2.8 to 3.5 of 5).

Esomonye et al., JACC Case Rep 2026

Where learning breaks down, and what changes

What gets in the way	What CoroSim does
ECGs are memorized as patterns, cut off from anatomy Students say past teaching "focused on memorizing traces" (Cardoso Pinto 2024)	Every tracing comes from a lesion in a 3D coronary tree, and the leads are drawn as axes, so learners see why ST rises or falls
Physiology is taught as static curves Misconceptions persist through training (Badenhorst 2021)	One closed loop: lesion, ischemia, ECG, pressure and reflexes, beat by beat
STEMI and shock decisions are rarely practiced	Learners treat while the muscle dies; every drug and device has its real cost, and warnings name the harm
Simulation time is scarce and tied to a lab	Free in any browser, at any hour; a link opens a specific case

Block an artery and watch the patient respond

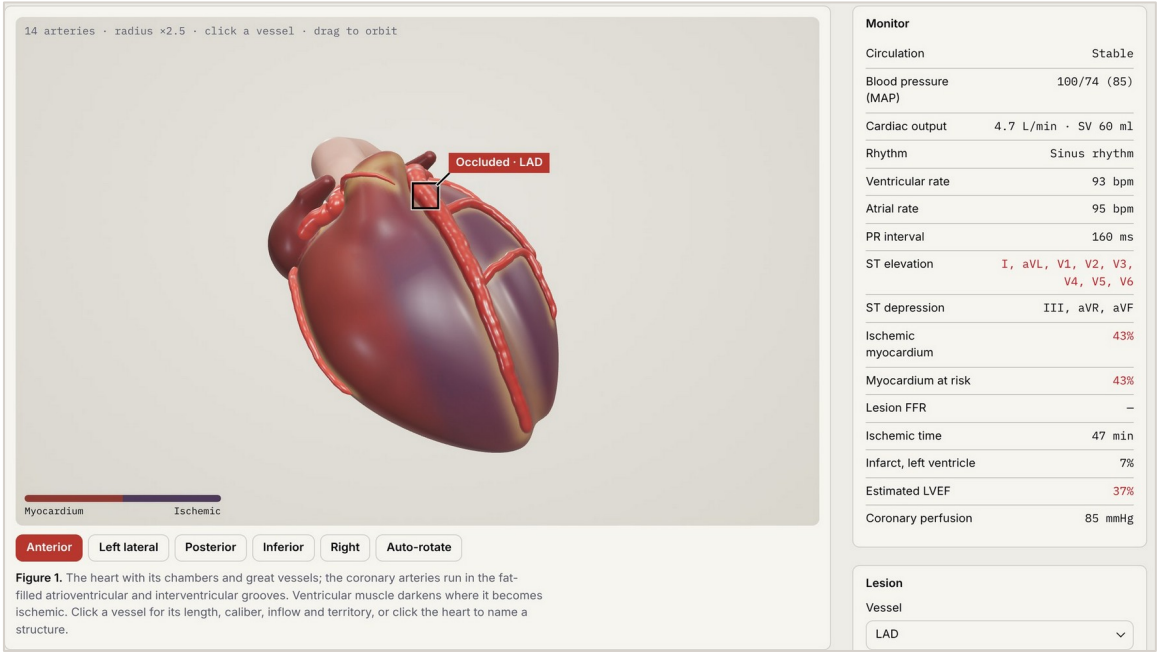


Figure 2. The simulator during a proximal LAD occlusion.

- Place a lesion on any of 14 arteries, across nine anatomic variants
- Ischemia, the 12-lead, rhythm, AV block and blood pressure all follow from the physics
- The infarct grows in clinical minutes and lowers LVEF until reperfusion

Read the 12-lead, then treat the patient

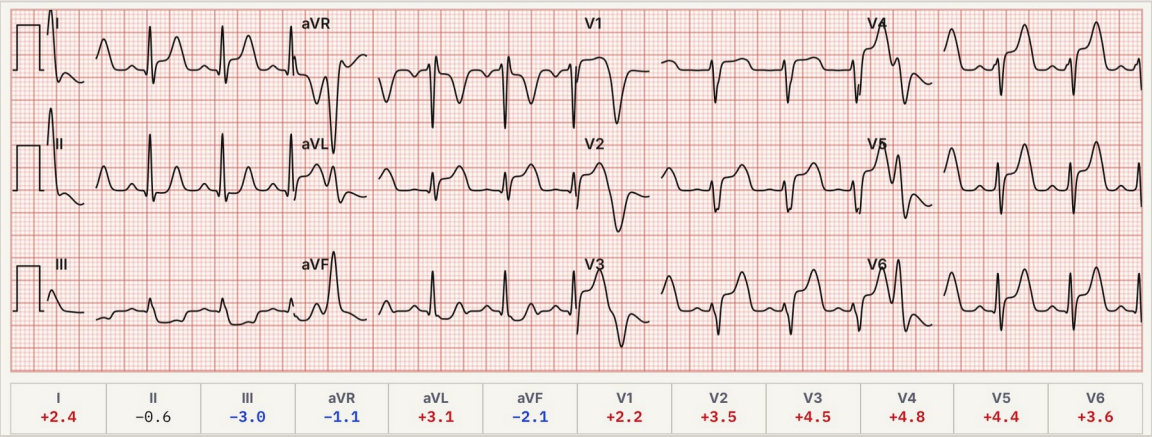


Figure 3. The simulator's own 12-lead: ST up in I, aVL and V1–V6, down in III and aVF.

- ST shifts point to the blocked artery
- Each drug and device carries its real cost
- PCI, lysis and CABG take their real time

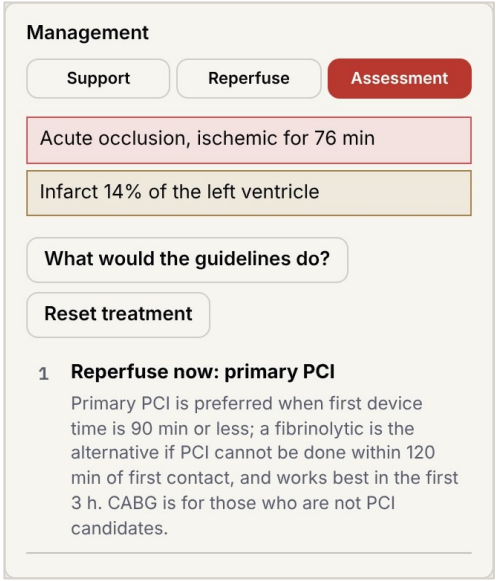


Figure 4. Assessment tab.

A daily hidden-lesion case, the same for everyone

- The lesion is hidden: read the tracing and monitor, then click where it is
- Points for accuracy along the vessel tree and for speed, then the RV call
- Then manage the case: points for muscle saved and time out of shock, minus harm

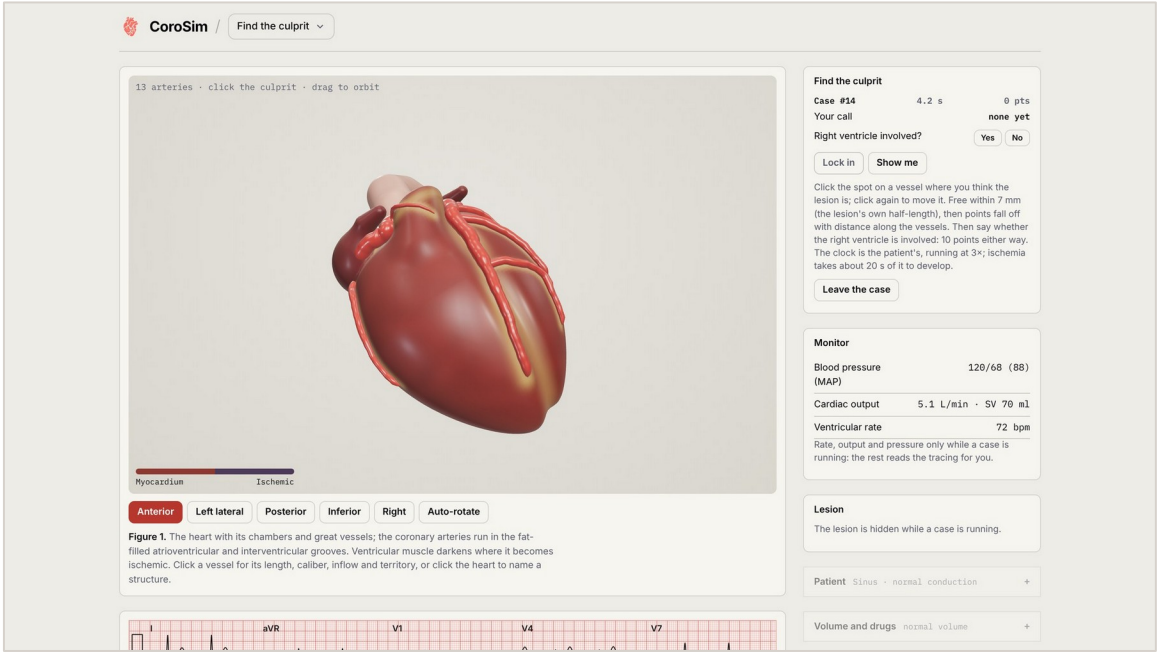


Figure 5. Find the culprit, case 14, mid-case.

More than 50 tracings, each drawn through a 3D heart

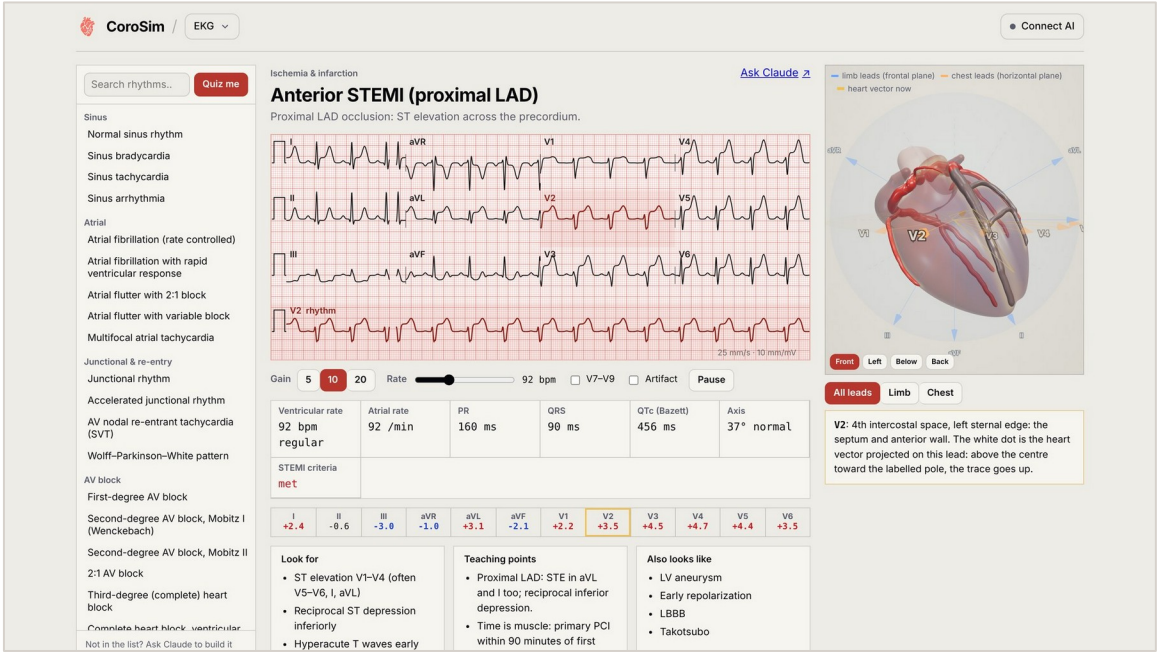


Figure 6. The EKG page, anterior STEMI preset.

- Rhythms, blocks, ischemia patterns and STEMI mimics, with teaching points
- Each lead is an axis through the heart, so learners see why it goes up or down
- Intervals, axis and ST are measured from the waveform; a quiz mode hides the name

A virtual cath lab on the same patient

- Angiography from any C-arm angle; dye moves at each vessel's solved flow
- A pressure-wire pullback reads Pd/Pa and FFR
- PCI, lysis and CABG show the vessel before and after

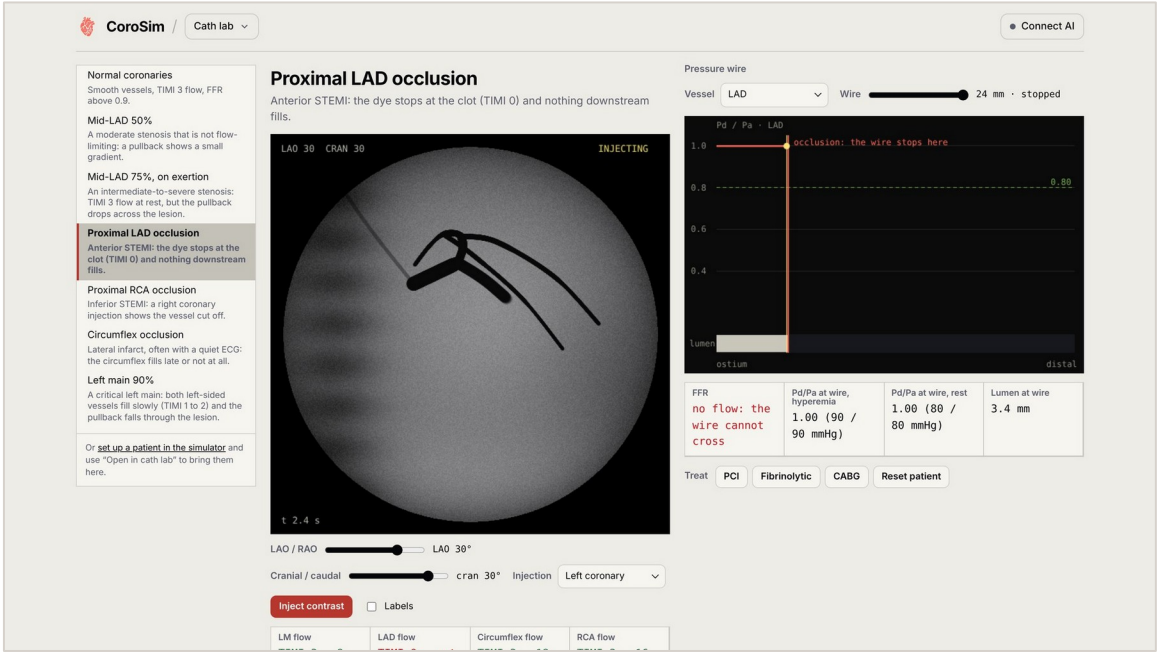


Figure 7. The cath lab, proximal LAD occlusion: TIMI 0 flow.

Valve and congenital lesions on a closed-loop heart

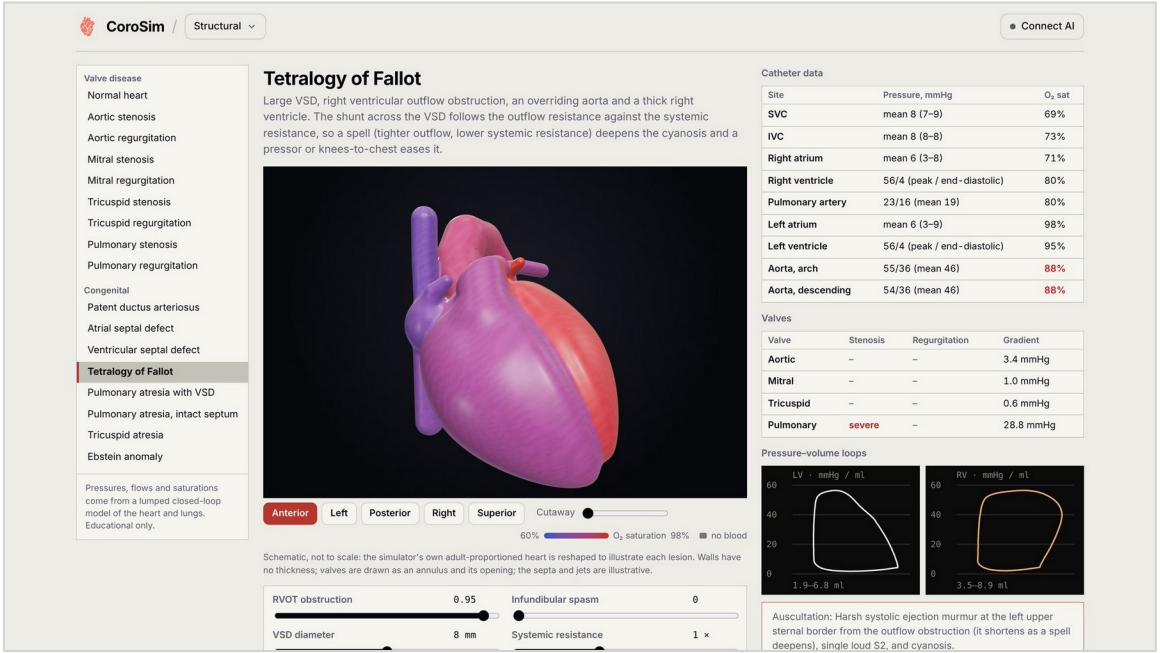


Figure 8. The structural page, tetralogy of Fallot.

- Eight valve lesions and eight congenital defects, each with severity sliders
- Pressures, saturations, gradients, Qp:Qs and pressure–volume loops update live
- Auscultation findings and oxygen saturation shown on the 3D heart

From probe position to the echo loop

- Probe and imaging plane on the 3D heart beside a real TTE loop
- Parasternal long axis, short axis and subcostal views
- Probe placement, index mark and what to look for

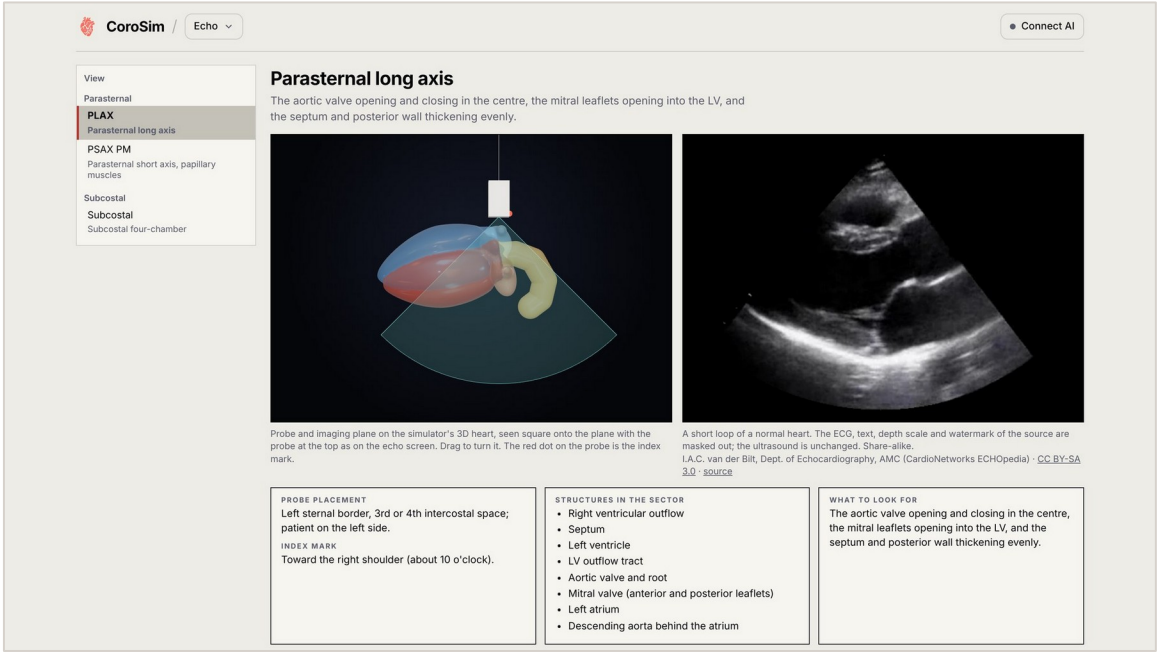


Figure 9. The echo atlas, parasternal long axis.

Six skills, and the pages that practice them

	Simulator	Find the culprit	EKG	Cath lab	Structural	Echo
Reading the ECG and localizing the culprit	●	●	●			
Coronary anatomy and territories	●	●		○		
Ischemia and hemodynamic physiology	●	○		○	●	
STEMI and shock management	●	●		○		
Reading angiograms, FFR and echo views				●		●
Valve and congenital hemodynamics					●	○

● practiced directly ○ reinforced

Students start with rows 1–3; residents add the rest.

Built for medical students and IM residents

Medical students

First-year physiology through the clerkships

- Cardiovascular physiology block and small groups
- Coronary anatomy and territories
- How a blocked artery becomes an ST change
- Normal and abnormal rhythms

Internal medicine residents

Wards, CCU and the cardiology rotation

- Noon conference, ECG conference and morning report
- Culprit localization and STEMI mimics
- Reperfusion timing, pressors, inotropes and support devices
- Right-ventricular infarct pitfalls

Three levels of evaluation, starting with one class

1 Reaction

- Usability (System Usability Scale)
- Perceived realism
- Confidence reading ECGs, before and after

2 Learning

- Pre/post test: culprit localization from 12-leads
- Rhythm identification
- Core cardiovascular concept questions

3 Behavior

- Performance across repeated daily cases
- Location error, time to call, harm avoided
- Course ECG and physiology exam items

Planned study

First-year medical students, RSOM Class of 2030

CoroSim is being planned for study among first-year students in RSOM's Class of 2030, measuring reaction, learning and behavior as above. In-app data already scores each case: location error, time to call, the RV call, muscle saved against immediate PCI, time in shock and harmful actions.

Why CoroSim matters in the wider training picture

- **Cause and effect, made changeable**

Cardiology is learned by seeing cause and effect; CoroSim makes the cause something a learner can change

- **One model from physiology to residency**

The same heart runs from the first-year physiology block to the CCU, so learners build one mental model and keep it

- **Practice of rare, high-stakes events**

Unlimited low-stakes repetitions of STEMI, shock and heart block, which rotations cannot guarantee

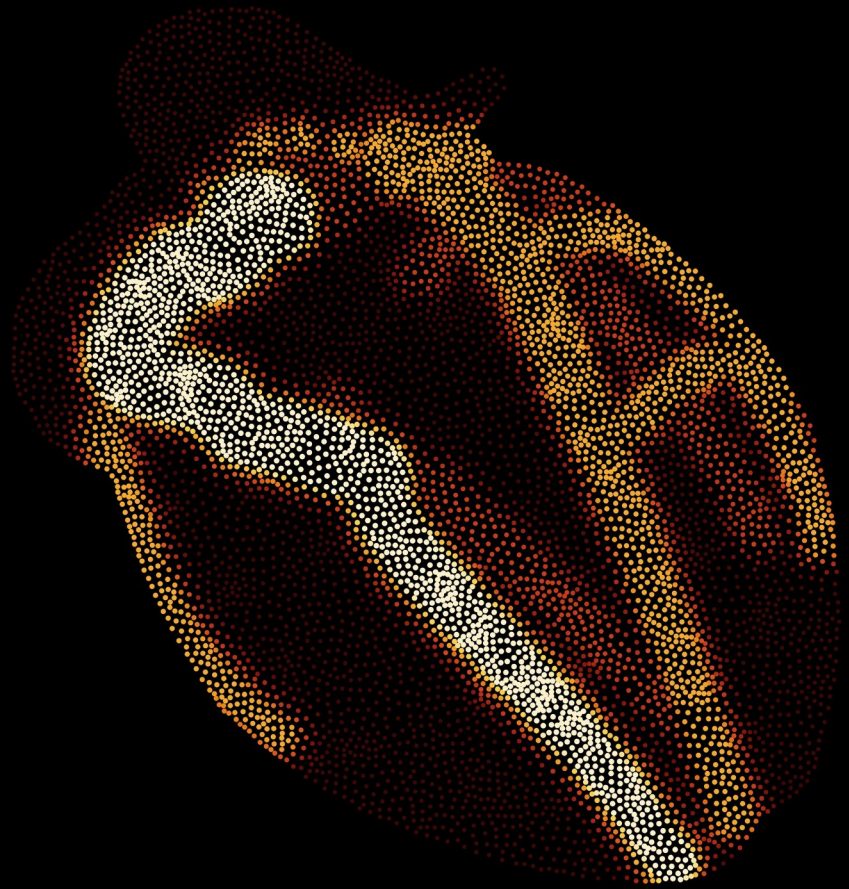
- **Complements existing simulation**

Comes before mannequin and ACLS sessions, so lab time goes to hands-on skills and teamwork

Fits into the teaching that already happens

Opens anywhere	Laptop, phone or the lecture-hall projector; no install, account or license.
Fits sessions already on the calendar	Physiology small groups, flipped pre-work, clerkship ECG teaching, noon conference and morning report.
Faculty assign exact cases	A link opens a set patient, tracing or cath case for a lecture, LMS module or exam.
An AI tutor on request	Connect AI lets Claude or ChatGPT run Socratic rounds on the learner's own patient.
A daily habit	One Find the culprit case a day. A daily ECG quiz for students held 81% participation over six weeks (Moulin 2025).

Open about its limits: every model constant is tagged literature-backed or assumed, more than 280 automated tests run on each change, and it is for education only. Validation against patient data is the next step.



Time is muscle.

Practice is free.

corosim.com

Educational simulation; not for clinical decisions.

Sources

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8. Image credits: echo loops, I.A.C. van der Bilt, AMC (CardioNetworks ECHOpedia), CC BY-SA 3.0; cath-lab heart mesh, Rodero C, et al. PLoS Comput Biol. 2021;17(4):e1008851, CC BY 4.0.